

Line Segment Intersection

1 Problem Definition

Segment Intersection

Given two line segments

$$\overline{P_1P_2} \quad \text{and} \quad \overline{P_3P_4},$$

decide whether the two segments intersect.

The output is a Boolean answer:

`true` or `false`.

Two segments intersect if they share at least one point. This includes:

- a proper crossing;
- touching at an endpoint;
- overlapping along a collinear interval.

2 Line Segment as a Convex Combination

Line Segment

For two endpoints

$$A = (a_x, a_y), \quad B = (b_x, b_y),$$

the segment $\overline{AB}$ is the set of all points

$$X = \alpha A + (1 - \alpha)B, \quad 0 \leq \alpha \leq 1.$$

The condition $0 \leq \alpha \leq 1$ means X lies between A and B , not outside the segment.

3 Orientation Review

For three points

$$A = (a_x, a_y), \quad B = (b_x, b_y), \quad C = (c_x, c_y),$$

the orientation test is

$$\text{orient}(A, B, C) = (b_x - a_x)(c_y - a_y) - (b_y - a_y)(c_x - a_x).$$

Value	Meaning
$\text{orient}(A, B, C) > 0$	$A \rightarrow B \rightarrow C$ makes a left turn.
$\text{orient}(A, B, C) < 0$	$A \rightarrow B \rightarrow C$ makes a right turn.
$\text{orient}(A, B, C) = 0$	A, B, C are collinear.

Sign Convention

These notes use

$$\text{orient}(A, B, C) = \overrightarrow{AB} \times \overrightarrow{AC}.$$

Some textbooks define a procedure called DIRECTION using the reversed cross-product order. That reverses the sign. In this note, always use the sign table above.

4 Main Idea: Straddling

Two segments properly intersect when each segment crosses the infinite line containing the other segment.

For

$$\overline{P_1P_2} \quad \text{and} \quad \overline{P_3P_4},$$

compute four orientation values:

$$\begin{aligned} d_1 &= \text{orient}(P_3, P_4, P_1), & d_2 &= \text{orient}(P_3, P_4, P_2), \\ d_3 &= \text{orient}(P_1, P_2, P_3), & d_4 &= \text{orient}(P_1, P_2, P_4). \end{aligned}$$

General Intersection Rule

$$d_1d_2 < 0 \quad \text{and} \quad d_3d_4 < 0 \quad \implies \quad \text{segments properly intersect.}$$

The condition $d_1d_2 < 0$ means P_1 and P_2 are on opposite sides of the line through P_3P_4 . The condition $d_3d_4 < 0$ means P_3 and P_4 are on opposite sides of the line through P_1P_2 .

5 Boundary Case: On Segment

If an orientation value is zero, then three points are collinear. But collinearity alone is not enough. The point must also lie between the segment endpoints.

On-Segment Test

Assume A, B, C are collinear. Then C lies on segment $\overline{AB}$ if

$$\min(a_x, b_x) \leq c_x \leq \max(a_x, b_x)$$

and

$$\min(a_y, b_y) \leq c_y \leq \max(a_y, b_y).$$

6 Complete Algorithm

Segments Intersect

```
SegmentsIntersect(p1, p2, p3, p4):
    d1 = orientation(p3, p4, p1)
    d2 = orientation(p3, p4, p2)
    d3 = orientation(p1, p2, p3)
```

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d4 = orientation(p1, p2, p4)

if d1 * d2 < 0 and d3 * d4 < 0:
    return true

if d1 == 0 and OnSegment(p3, p4, p1):
    return true

if d2 == 0 and OnSegment(p3, p4, p2):
    return true

if d3 == 0 and OnSegment(p1, p2, p3):
    return true

if d4 == 0 and OnSegment(p1, p2, p4):
    return true

return false

OnSegment(a, b, c):
    return min(ax, bx) <= cx <= max(ax, bx)
           and min(ay, by) <= cy <= max(ay, by)

```

7 Example 1: Proper Crossing

Problem

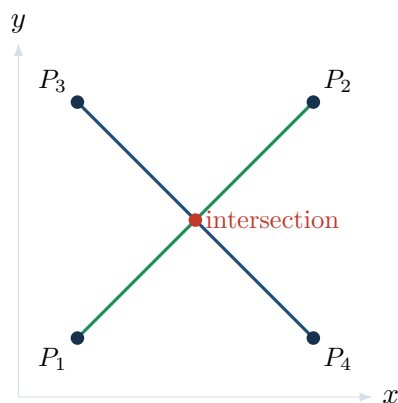
Test whether

$$P_1 = (1, 1), \quad P_2 = (5, 5)$$

and

$$P_3 = (1, 5), \quad P_4 = (5, 1)$$

intersect.



Compute:

$$d_1 = \text{orient}(P_3, P_4, P_1) = (5 - 1)(1 - 5) - (1 - 5)(1 - 1) = 4(-4) - (-4)(0) = -16.$$

$$d_2 = \text{orient}(P_3, P_4, P_2) = (5 - 1)(5 - 5) - (1 - 5)(5 - 1) = 4(0) - (-4)(4) = 16.$$

$$d_3 = \text{orient}(P_1, P_2, P_3) = (5-1)(5-1) - (5-1)(1-1) = 4(4) - 4(0) = 16.$$

$$d_4 = \text{orient}(P_1, P_2, P_4) = (5-1)(1-1) - (5-1)(5-1) = 4(0) - 4(4) = -16.$$

Now:

$$d_1 d_2 = (-16)(16) < 0, \quad d_3 d_4 = (16)(-16) < 0.$$

Answer

The two segments properly intersect.

8 Example 2: No Intersection

Problem

Test whether

$$P_1 = (1, 1), \quad P_2 = (5, 1)$$

and

$$P_3 = (2, 3), \quad P_4 = (4, 4)$$

intersect.

Use the line through $P_1 P_2$:

$$d_3 = \text{orient}(P_1, P_2, P_3) = (5-1)(3-1) - (1-1)(2-1) = 8.$$

$$d_4 = \text{orient}(P_1, P_2, P_4) = (5-1)(4-1) - (1-1)(4-1) = 12.$$

Both d_3 and d_4 are positive, so P_3 and P_4 are on the same side of the line through $P_1 P_2$. Therefore the second segment does not straddle the first line.

Answer

The two segments do not intersect.

9 Example 3: Endpoint on Segment

Problem

Test whether

$$P_1 = (1, 1), \quad P_2 = (5, 1)$$

and

$$P_3 = (3, 1), \quad P_4 = (3, 4)$$

intersect.

Point $P_3 = (3, 1)$ lies on the horizontal segment $\overline{P_1 P_2}$. Verify this using orientation:

$$d_3 = \text{orient}(P_1, P_2, P_3) = (5-1)(1-1) - (1-1)(3-1) = 0.$$

Since $d_3 = 0$, the points P_1, P_2, P_3 are collinear. Now apply the on-segment test:

$$\min(1, 5) \leq 3 \leq \max(1, 5), \quad \min(1, 1) \leq 1 \leq \max(1, 1).$$

That is,

$$1 \leq 3 \leq 5, \quad 1 \leq 1 \leq 1.$$

Both are true, so P_3 lies on $\overline{P_1P_2}$.

Answer

The two segments intersect at $P_3 = (3, 1)$.

10 Example 4: Collinear but Disjoint

Problem

Test whether

$$P_1 = (1, 1), \quad P_2 = (3, 1)$$

and

$$P_3 = (4, 1), \quad P_4 = (6, 1)$$

intersect.

All four points lie on the same horizontal line. For example:

$$\text{orient}(P_1, P_2, P_3) = (3 - 1)(1 - 1) - (1 - 1)(4 - 1) = 0.$$

But $P_3 = (4, 1)$ is not on segment $\overline{P_1P_2}$, because

$$\min(1, 3) \leq 4 \leq \max(1, 3)$$

becomes

$$1 \leq 4 \leq 3,$$

which is false.

Also, $P_2 = (3, 1)$ is not on segment $\overline{P_3P_4}$, because

$$4 \leq 3 \leq 6$$

is false.

Answer

The segments are collinear but disjoint.

11 Example 5: Overlapping Collinear Segments

Problem

Test whether

$$P_1 = (1, 1), \quad P_2 = (5, 1)$$

and

$$P_3 = (3, 1), \quad P_4 = (7, 1)$$

intersect.

The segments are collinear:

$$\text{orient}(P_1, P_2, P_3) = 0, \quad \text{orient}(P_1, P_2, P_4) = 0.$$

Now check whether an endpoint of one segment lies on the other segment. For $P_3 = (3, 1)$:

$$\min(1, 5) \leq 3 \leq \max(1, 5), \quad \min(1, 1) \leq 1 \leq \max(1, 1).$$

So:

$$1 \leq 3 \leq 5, \quad 1 \leq 1 \leq 1.$$

Both are true, so P_3 lies on $\overline{P_1P_2}$. Therefore the segments overlap.

Answer

The segments intersect along the overlap from $(3, 1)$ to $(5, 1)$.

12 Complexity

The algorithm computes four orientation values and a constant number of range checks.

Time Complexity

$$\mathcal{O}(1)$$

13 Final Review

Segment Intersection Summary

$d_1 = \text{orient}(P_3, P_4, P_1), \quad d_2 = \text{orient}(P_3, P_4, P_2).$
 $d_3 = \text{orient}(P_1, P_2, P_3), \quad d_4 = \text{orient}(P_1, P_2, P_4).$
 $d_1d_2 < 0$ and $d_3d_4 < 0 \Rightarrow$ proper intersection.
 $d_i = 0 \Rightarrow$ use the on-segment boundary check.
The full test runs in $\mathcal{O}(1)$.